

1. Project Overview

The platform allows lab owners and technicians to upload, track, and manage NABL compliance documents based on predefined rules configured by Admin.

The system includes Machine Types, Machine Instances, Document Templates, Calibration Tracking, Version Control, Clause Mapping, and a Compliance Dashboard.

Admins define compliance rules, and labs operate within those rules to meet NABL requirements.

2. User Roles & Access

Super Admin

- Full access to all labs, users, machine types, document templates, machine instances, approvals, audit logs, and system settings.

Admin

- Access to assigned labs.
- Can manage users, machine types, document templates, machine instances, approvals, notifications, and compliance tracking.

Lab Owner

- Manages lab-specific items including machine instances, document uploads, calibration information, and reports.
- Can view dashboards for all labs under their account and switch to technician-level view.

Lab Technician

- Responsible for uploading required documents for assigned labs and viewing assigned machines, reminders, and admin feedback.
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3. Core Features & Functionalities

3.1 Authentication Module

3.1.1 User Authentication

- Login for all user roles.
 - Forgot password with email verification.
 - Profile management for updating personal details and changing passwords.
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3.2 Admin Module

3.2.1 User Management

Admin can:

- Add, edit, deactivate, delete Lab Owners and Lab Technicians.
 - Assign users to specific labs.
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3.2.2 Machine Types Management (Admin Only)

Defines standard machine categories used across all labs.

Fields

- Machine Type Name
- Category
- Default Calibration Frequency
- Default Maintenance Frequency
- Status (Active/Inactive)
- Optional Notes
- **Required Documents (Multi-select from Document Templates)**

Purpose

- Machine Types act as the master compliance definition.
- Required document templates attached to Machine Types automatically apply to machine instances.

Admin Capabilities

- Create and maintain machine types.
- Map document templates to machine types.
- Update frequencies and rules for each type.
- Activate/deactivate machine types.

Lab Owners and Technicians can only view Machine Types; they cannot edit them.

3.2.3 Document Template Management (Admin Only)

A Document Template defines the type of document required for NABL compliance.

Fields

- Document Name
- Description
- Frequency (Daily, Weekly, Monthly, Quarterly, Yearly, One-Time)
- Allowed File Types
- NABL Clause Mapping
- Optional Help Content (text, video, or PDF)
- Status (Active/Inactive)

Important Architecture Rule

Document Templates **do not** select Machine Types.
Machine Types select Document Templates.

3.2.4 Machine Instance Management (Lab Owner + Admin)

Machine Instances represent actual equipment used in a specific lab.

Fields

- Machine Type (selected from Machine Type list)
- Machine Nickname (optional)
- Model
- Serial Number
- Calibration Due Date
- Maintenance Due Date
- Status (Active, Under Maintenance, Out of Service)
- Notes

Lab Owner Capabilities

- Add/edit/delete machine instances.
- Maintain calibration and maintenance due dates.
- Upload calibration certificates via document upload workflow.

Admin Capabilities

- Add/edit machine instances on behalf of lab owners as needed.
 - Verify machine entries.
 - Review calibration status across labs.
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3.2.5 Auto-Generation of Required Documents (Mandatory)

When a Machine Instance is created:

- System reads the Machine Type.
- It identifies all linked Document Templates.
- It auto-generates required document entries for that machine instance based on frequency.

This gives the lab a clear checklist of what they must upload over time.

3.2.6 Document Upload and Approval Workflow

Lab Technician

- Selects required document entry.
- Uploads file through the standard upload modal.
- Selects applicable date.
- Adds comments and submits.

Admin

- Receives upload notification.
- Reviews the uploaded document.
- Approves or rejects with feedback.
- Technician resubmits if needed.
- Full history of uploads per document is maintained.

All document types, including **Calibration Certificates**, use this same upload flow.

3.2.7 Version Control (Mandatory)

- Each approved upload is stored as a new version (v1, v2, v3).
 - Admin can view previous versions and revert if needed.
 - The system displays the current active version to Lab Owners and Technicians.
 - Version control ensures NABL compliance and audit readiness.
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3.2.8 Notification Module

Admin notifications:

- New document uploads
- Pending approvals
- Calibration approaching due
- Ability to send broadcast messages to Lab Owners

Lab Owner/Technician notifications:

- Pending document uploads
- Document rejection feedback
- Calibration due alerts
- Frequency-based reminders

Delivery via:

- Internal notifications
 - Email notifications
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3.2.9 Help Content Management

Admin:

- Add help articles, images, videos, and document guides.
- Assign help content to specific document templates.

Lab Owners/Technicians:

- Access help through an eye icon located next to each required document.
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3.3 Lab Owner / Technician Module

3.3.1 Dashboard & Reports

Displays:

- Total required documents
- Pending uploads
- Overdue uploads
- Machine list

- Calibration due today / this week / overdue
- Document compliance by frequency
- Clause-wise grouping of document requirements
- Overall compliance score preview

Lab Owner can:

- Switch between labs.
- Open technician-level dashboard for each lab.

Technicians see only their assigned lab.

3.3.2 Document Upload Process (Single Standard Workflow)

Applies to all document templates including:

- Daily logs
- Weekly checklists
- Monthly verification
- Preventive maintenance
- Calibration certificates
- SOP acknowledgement
- Any other compliance document

Technician:

- Uploads file
- Selects applicable date
- Adds comments
- Submits for approval

Admin:

- Approves or declines
 - Provides feedback
 - Version is created upon approval
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3.3.3 Machine Instance View

Lab Owner/Technician can:

- View machine list
- View calibration and maintenance due dates
- View required documents associated with each machine

- Quickly navigate to upload required documents
(All uploads use the standard Document Upload workflow outlined in 3.3.2)
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3.3.4 Calibration Module (Tracking Only)

Calibration Certificate is a **Document Template**, so uploads happen through the standard flow.

Calibration Module provides:

- Dashboard of calibration due dates
- Upcoming and overdue calibration alerts
- Machine compliance status
- Filter to view calibration-related documents for each machine
- Quick link to upload the document (opens standard upload modal)
- Admin view of calibration compliance across all labs

No separate upload system exists for calibration.

3.3.5 Help Content Access

Lab Owners and Technicians:

- Access help content via eye icon next to each document requirement.
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3.3.6 Notifications

- Pending upload notifications
 - Document approval/rejection notifications
 - Calibration reminders
 - Compliance frequency reminders
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4. Storage & Hosting

- All documents stored in AWS S3.
 - Metadata stored in a secure database.
 - Web application hosted on a secure server.
 - Strict role-based access control implemented.
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5. Audit Log & History Tracking

Audit Log (Admin, Super Admin, Lab Owner)

Captures:

- Document uploads
- Approvals and rejections
- Version changes
- Machine instance updates
- Calibration date updates
- User actions with timestamps and IP addresses

History Log (All Roles)

Shows:

- Document upload history
 - Version history
 - Status changes
 - Rejection comments
 - Machine instance history
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6. Modules Excluded from Initial Demo (Phase 2+)

- PTP/EQAS tracking
- NCR & CAPA module
- Internal Audit module
- Staff competency and training
- WhatsApp ingestion
- Email ingestion
- LIMS/instrument integrations
- Automated NABL audit package generation